

# Antonio De Leon

Santa Cruz, CA | jaguir26@ucsc.edu | antonio-de-leon.com | github.com/AntonioAPDL | linkedin.com/in/antonioapdl

## Professional Summary

Statistics Ph.D. candidate and applied quantitative researcher with experience in probabilistic forecasting, statistical learning, Bayesian computation, and reproducible model evaluation. Builds analytical workflows in Python, R, SQL, and C++ across financial, sensor, image, and environmental data, from data construction through diagnostics, software, and communication.

## Education

**Ph.D. in Statistics, University of California, Santa Cruz** Expected December 2026

GPA: 3.82/4.00; candidacy completed. Research: probabilistic forecasting, model evaluation, and statistical software.

**M.Sc. in Economic Theory, ITAM** 2018–2020

GPA: 96.7/100.

**B.Sc. in Applied Mathematics, ITAM** 2014–2018

GPA: 97.1/100.

*Relevant coursework and training:* probability, statistical inference, regression, statistical learning, Bayesian inference, time series, stochastic processes, decision theory, causal inference, and impact evaluation.

## Technical and Quantitative Skills

**Programming:** Python, R, SQL, C++, Bash, Git, LaTeX.

**Statistical and machine-learning methods:** probability, regression, Bayesian inference, time-series and state-space models, statistical learning, Monte Carlo methods, variational inference, experimental design, causal inference and impact-evaluation training, model diagnostics, calibration, and proper scoring.

**Systems and tools:** Linux, AWS (EC2, EMR, S3), Stan, Rcpp, CRAN, GitHub Actions, Tableau, DBeaver.

**Languages:** Spanish, English, German.

## Professional Experience

**Delos Financial Technologies, San Rafael, CA (Remote)** May 2025–Sep. 2025

*Quantitative Researcher (Contract); Python, Git, Bash, C++, AWS, Stan*

- Built Python model-evaluation workflows for residual diagnostics, goodness-of-fit analysis, cohort summaries, and backtest review across financial-modeling experiments.
- Implemented baseline and extended forecasting specifications, automated AWS experiment runs, and standardized versioned outputs for reproducible model comparison.

**UCSC Statistics Department, Santa Cruz, CA** Feb. 2024–Present

*Computer Systems Coordinator; Linux, Bash, Python/R, Networking, User Support*

- Administer Linux research infrastructure for faculty and graduate students, including software stacks, storage, upgrades, access, and incident response.
- Write automation scripts and documentation for onboarding, resource usage, and reproducible computing; coordinate with central IT on research infrastructure planning.

**NeatLeaf Inc., Scotts Valley, CA** May 2021–Sep. 2022

*Data Analyst; Python, SQL, Git, Bash, Linux, C++, DBeaver*

- Built Python and SQL pipelines for greenhouse robot and sensor telemetry used in operational monitoring and downstream analysis.
- Analyzed spatiotemporal microclimate anomalies and robot/sensor calibration logs to improve data-quality review and system reliability.

**Banco de México (BANXICO), Mexico City, Mexico** Aug. 2018–Dec. 2019

*Data Analyst; Python, R, SQL, Git*

- Built Python/R/SQL pipelines for banknote image and operational datasets used in monitoring, anomaly analysis, and model development.
- Developed anomaly-classification diagnostics and financial-volume forecasting prototypes for operational planning and review.

## Selected Research and Software

**exdqIm – CRAN R package v1.1.0** 2026

*Software article submitted to the Journal of Statistical Software*

- Co-developed a public package for Bayesian dynamic and static quantile models, including MCMC and variational inference, forecasting, diagnostics, tests, documentation, and reproducible examples.
- Added evaluation utilities and selected C++ paths, with R reference implementations for validation and portability.

## Bayesian Quantile Deep Echo State Networks

2024–Present

*Working paper on nonlinear time-series forecasting*

- Developed probabilistic time-series models using fixed nonlinear recurrent features, regularization, MCMC, and variational inference.
- Designed simulation and rolling held-out studies, including a selected-panel electricity-price forecasting comparison with 42 region/fold evaluations.

## Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products

2021–Present

*arXiv preprint; manuscript submitted to Environmetrics*

- Integrated USGS observations, retrospective products, ensemble forecasts, and covariates from multiple agencies into origin-specific, version-consistent datasets.
- Evaluated probabilistic forecasts across five origins and seven fitted quantile levels using CRPS and quantile loss, keeping post-origin observations reserved for verification.

## Selected Publications

---

**De Leon, A., Barata, R., Prado, R., & Sansó, B.**

*exdq1m: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models.*

Submitted to the *Journal of Statistical Software*; CRAN R package v1.1.0; arXiv:2607.22760.

**De Leon, A., Prado, R., & Sansó, B.**

*Bayesian Quantile Deep Echo State Networks.*

Working paper on probabilistic time-series modeling with fixed reservoir features, MCMC, variational inference, and held-out forecast evaluation.

**De Leon, A., Prado, R., & Sansó, B.**

*Mean-Tilted Intervals: A Generalized-Bayes Approach to Fixed-Content and Tolerance Intervals.*

arXiv preprint on MTI fixed-content and tolerance-interval targets, generalized-Bayes computation, and calibrated tolerance actions; arXiv:2607.26098.

**De Leon, A., Prado, R., & Sansó, B.**

*Mean-Tilted Intervals: Regression and Dynamic Models.*

Submitted manuscript on MTI regression and dynamic state-space extensions; public identifier pending.

**De Leon, A., Prado, R., & Sansó, B.**

*Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products.*

arXiv preprint; manuscript submitted to *Environmetrics*; arXiv:2608.11222.

**De Leon, A., & Lobato, I. N.**

*Evidence of non-fundamentality in OECD capital stocks.*

*Empirical Economics* 67, 761–772 (2024).

## Awards, Teaching, and Service

---

**Awards:** Best Poster Prize, ISBA 2026 World Meeting; EnviBayes Student Paper Competition Winner, 2026; ISBA 2026 Junior Researcher Travel Award; UC Regents Fellowship, UCSC Statistics, 2021; Baillères Fellowship, ITAM, 2014–2020.

**Teaching and mentoring:** Graduate Student Instructor for *Data Visualization*; Teaching Assistant for *Probability Theory*, *Classical and Bayesian Inference*, *Statistics*, and related courses; ASA DataFest mentor for student teams in modeling, visualization, and presentation.

**Service and training:** TA Wiki maintainer and workshop contributor for Git, GitHub, collaborative documentation, and departmental teaching resources; contributed to an anonymous, IRB-exempt TA Wiki survey and follow-up resource improvements.